

**GRAPH THEORY:
HOMEWORK 6
BOOK SECTION 3.1 PART II.**

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1. Prove or disprove: Every tree has at most one perfect matching.

Proof.

□

2. Prove that every maximal matching in a graph G has at least $\alpha'(G)/2$ edges.

Proof.

□

3. Prove that if G is an X, Y -bigraph such that

$$|N(S)| > |S| \quad \text{whenever } \emptyset \neq S \subsetneq X,$$

then every edge of G belongs to some matching that saturates X .

Proof.

□

4. Use the König–Egerváry Theorem to prove that every bipartite graph G has a matching of size at least $e(G)/\Delta(G)$. Use this to conclude that every subgraph of $K_{n,n}$ with more than $(k-1)n$ edges has a matching of size at least k .

Proof.

□

5. Let G be an X, Y -bigraph with vertex cover Q , and let H be the induced $(X-Q), (Q \cap Y)$ -sub-bigraph of G . Prove that if no matching in H saturates $Q \cap Y = Q - X$, then G has a vertex cover that is smaller than Q .

Proof.

□